Conserve mechanical energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Track kinetic and potential energy when nonconservative work is negligible.

SUCCESS CRITERIA - I can...

1. Identify the energy system and assumptions.
2. Keep the total energy consistent between positions.
3. Explain how height and speed trade off.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “mechanical energy” and “conserved” to explain your model or result. Sentence starters are optional.

mechanical energy: total kinetic and potential energy in a model

conserved: unchanged in total under the stated conditions

system: the object or objects chosen for analysis

01 Warm-up: bring back an earlier idea

Lift a 3 kg object by 5 m.
Use g=10 N/kg.
Find the change in gravitational potential energy.

Word help: potential energy = energy associated with the configuration of interacting objects

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $K_i + U_i = K_f + U_f$; $K = \frac{1}{2}mv^2$; $U = mgh$.
Mechanical energy conserved when only conservative interactions transfer energy. Use g = 10 N/kg near the surface.

Worked reasoning: A 1 kg object drops 5 m from rest with g=10 m/s² and no energy loss. It loses 50 J of potential energy and gains 50 J of kinetic energy, giving speed 10 m/s.

Your turn: Draw energy bars at the start and just before arrival.
Label the system and neglected losses.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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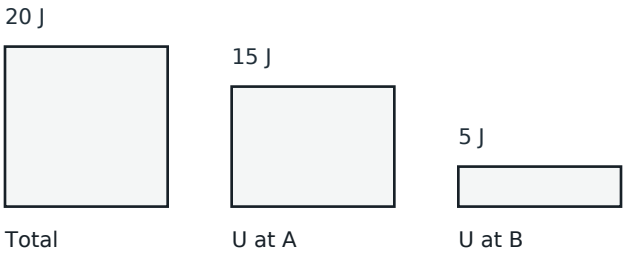
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A system has total mechanical energy 20 J. At one point U=15 J; at another U=5 J.
Find K at each point.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: 1 2 3

04 Apply the physics

A 1 kg ball is released from rest 2 m above the floor.
Use g = 10 N/kg; ignore air resistance. Complete an energy account from release to just before reaching the floor.
Find initial U and final K.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A cart loses mechanical energy on a rough track. Does total energy conservation fail?
Name the destination to include.

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

An impactor has 4 times the original mass at the same speed and stopping time.
Compare momentum and average stopping-force size.
Use before and after snapshots to organize the momentum account.

Word help: momentum = mass times velocity, including direction

Sentence starter: Both ... . They differ in ... because ... .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

An ideal system has 30 J total. At a point U=12 J, what is K?
Use the energy model to show the quantities you compare; label units and any unknowns.
Use g = 10 N/kg when needed.

Word help: ideal = using stated simplifying conditions

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two frictionless ramps start at the same height and end at the same lower height.
Compare final speeds and explain what need not be equal.
Use the energy model to show the quantities you compare; label units and any unknowns.
Use g = 10 N/kg when needed.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3